This can be written as

$$V_{\rm quad} = \frac{1}{2}\begin{bmatrix} \varphi \\ \varphi^* \end{bmatrix}^\dagger M_0^2 \begin{bmatrix} \varphi \\ \varphi^* \end{bmatrix}, \tag{27.4.16}$$

where M_0^2 is the block matrix

$$M_0^2 = \begin{bmatrix} \mathscr{M}^*\mathscr{M} + \sum_A (t_A\phi_0)(t_A\phi_0)^\dagger & \sum_A (t_A\phi_0)(t_A\phi_0)^{\rm T} \\ \sum_A (t_A\phi_0)^*(t_A\phi_0)^\dagger & \mathscr{M}\mathscr{M}^* + \sum_A (t_A\phi_0)^*(t_A\phi_0)^{\rm T} \end{bmatrix}. \tag{27.4.17}$$

Now we must look for the eigenvalues of this mass-squared matrix. Differentiating Eq. (27.4.12) with respect to ϕ_n gives

$$\sum_m \frac{\partial^2 f(\phi)}{\partial\phi_n\partial\phi_m}\Big(t_A\phi\Big)_m + \sum_m \frac{\partial f(\phi)}{\partial\phi_m}\Big(t_A\Big)_{mn} = 0\,. \tag{27.4.18}$$

But as we have seen, $\partial f(\phi)/\partial\phi_m$ vanishes at $\phi = \phi_0$, so by setting ϕ at this value in Eq. (27.4.18), we find

$$\sum_m \mathscr{M}_{nm}\,(t_A\phi_0)_m = 0\,. \tag{27.4.19}$$

It follows that

$$M_0^2 \begin{bmatrix} t_B\phi_0 \\ \pm(t_B\phi_0)^* \end{bmatrix} = \sum_A \Big(\phi_0^\dagger[t_A t_B \pm t_B t_A]\phi_0\Big) \begin{bmatrix} t_B\phi_0 \\ \pm(t_B\phi_0)^* \end{bmatrix}.$$

But the vanishing of D_A at $\phi = \phi_0$ and the global gauge invariance of ξ_A tell us that

$$\Big(\phi_0^\dagger[t_A\,,\,t_B]\phi_0\Big) = i\sum_C C_{ABC}\Big(\phi_0^\dagger t_C\phi_0\Big) = -i\Big(\phi_0^\dagger\phi_0\Big)\sum_C C_{ABC}\xi_C = 0\,. \tag{27.4.20}$$

The matrix (27.4.17) therefore has a pair of eigenvectors for each gauge symmetry

$$u = \begin{bmatrix} \sum_B c_B\, t_B\phi_0 \\ \sum_B c_B\,(t_B\phi_0)^* \end{bmatrix}, \qquad v = \begin{bmatrix} \sum_B c_B\, t_B\phi_0 \\ -\sum_B c_B\,(t_B\phi_0)^* \end{bmatrix}, \tag{27.4.21}$$

for which

$$M_0^2 u = \mu^2 u\,, \qquad M_0^2 v = 0\,, \tag{27.4.22}$$

where μ^2 and c_A are any real solutions of the eigenvalue problem*

$$\sum_B \Big(\phi_0^\dagger\{t_A\,,\,t_B\}\phi_0\Big)c_B = \mu^2 c_A\,, \tag{27.4.23}$$

* The presence of a factor 1/2 in Eq. (21.1.17) which does not appear in Eq. (27.4.23) is due to a difference in the way that the scalar fields are normalized.